Class Test 2

Electricity and Electronics EBN 111

10 May 2024

Test Information			
Maximum marks:	16	Full marks:	15
Duration of test:	30 min + (10 min for OCR)	Open/Closed book:	Closed
Total number of pages (this page included)		8	

Important

1. The examination regulations of the University of Pretoria apply.
2. Answer all questions on the OCR forms provided. The question number **01** in this question paper refers to numeric field 01 of the OCR form.
3. Read the instructions on the OCR form carefully.
4. Where applicable, answers must include units.
5. Final answers must be written as real numbers and not as fractions.
6. Use at least 3 significant numbers to write irrational numbers.

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Assistant Lecturers: S. Twala, L. Hurter, J. Nepembe

INSTRUCTIONS

Do step 0 in the first 5 minutes of the session. (-5 to 0 min):

Do step 1 in the first 30 minutes of the test. (0-30 min):

Do step 2 within the following 10 minutes. (30-40 min):

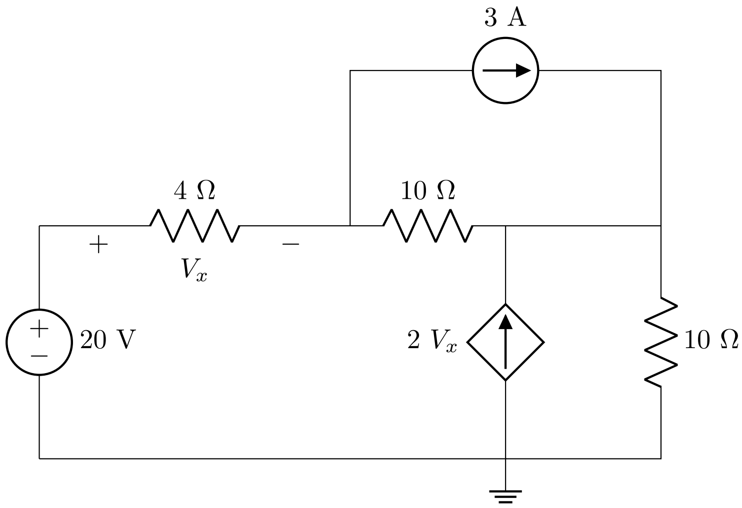
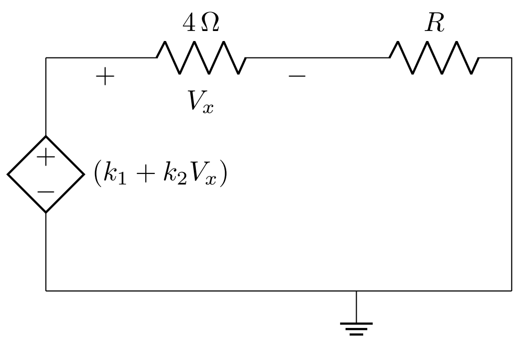
STEP 0 (-5 min to 0 min): Complete the front page of the OCR form, and also fill in your name and student number on additional pages of the OCR form.

STEP 1 (0 min to 30 min): Do your own calculations on the question paper. Write down the answers for numeric questions on the PRACTICE ANSWER SHEET. Be sure to include the units where applicable.

STEP 2 (30 min to 35 min): (only if numeric sheets are completed) Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2024EBN111C02**
- Use a mechanical pencil with 0.5 mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell. A negative sign is its own character.
- **Do not use commas!** Please make use of full stops as decimal points.

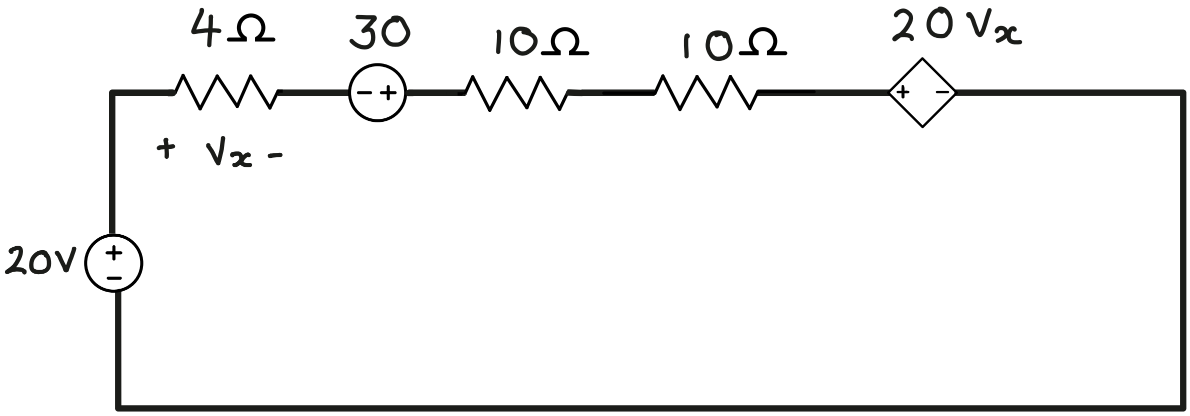
Section 1

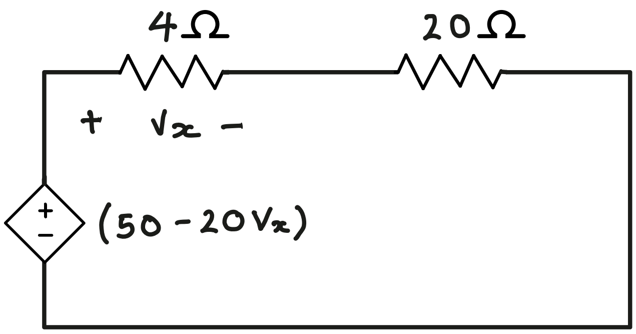
Questions 1 to 3 [4]
Use source transformation to convert Circuit A to Circuit B.
Circuit A: 
Circuit B: 

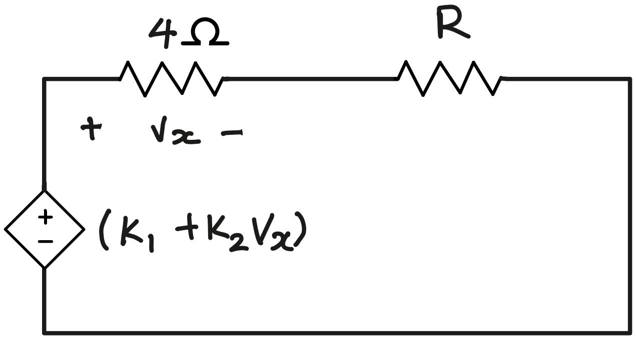
01	Find R [2]
02	Find k_1 [1]
03	Find k_2 [1]

memo on next page

This page is intentionally left open for rough work.



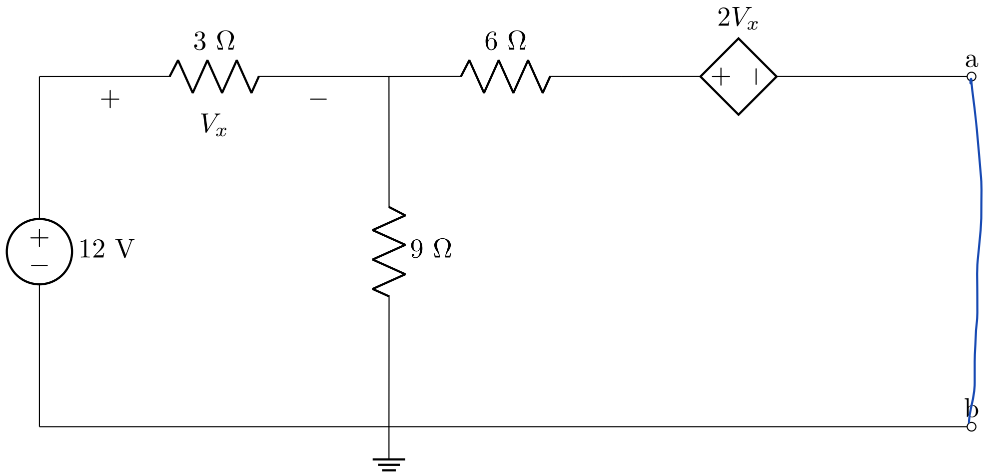




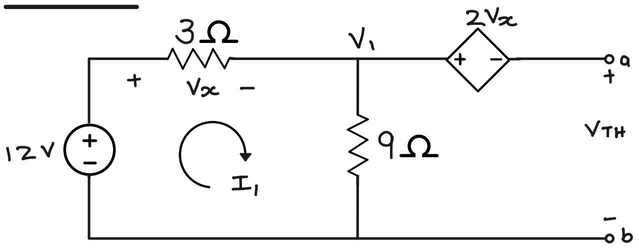
$\therefore$ $R = 20\Omega$

$K_1 = 50$

$K_2 = -20$

Questions 4 to 5 [4]	
Find the Thevenin equivalent voltage V_{Th} and resistance R_{Th} with respect to terminals $a - b$.	
	
04	V_{Th} [2] 3V
05	R_{Th} [2] 12.75Ω

For V_{TH}:



The 6 ohm resistor is removed since no current flows through it

KVL: $-12 + 3I_1 + 9I_1 = 0$

$\therefore I_1 = 1A$

$\therefore V_x = 3I_1 = 3(1) = 3V$

$\therefore 2V_x = 2(3) = 6V$

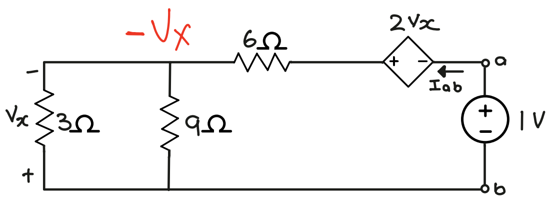
$V_1 = 9I_1 = 9(1) = 9V$

$\therefore V_1 - V_{TH} = 2V_x$

$\therefore 9 - V_{TH} = 6$

$\therefore V_{TH} = 3V$

For R_{TH}:



$$\frac{-V_x}{3} + \frac{-V_x}{9} + \frac{-V_x - 2V_x - 1}{6} = 0$$

$$\frac{V_x}{3} + \frac{V_x}{9} + \frac{V_x}{2} + \frac{1}{6} = 0$$

$$\frac{17}{18}V_x + \frac{1}{6} = 0$$

$$\Rightarrow 17V_x + 3 = 0, \quad V_x = -\frac{3}{17}$$

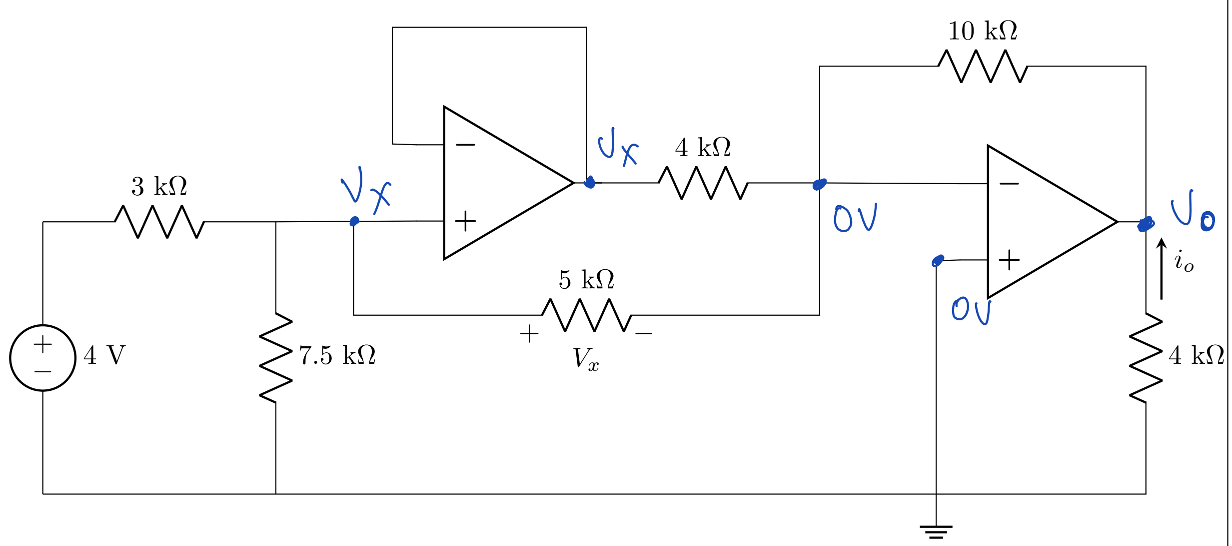
$$I_{ab} = \frac{1 + 2V_x - (-V_x)}{6} = \frac{1 + 3V_x}{6}$$

$$= \frac{1 - \frac{9}{17}}{6} = \frac{8}{17 \times 6} A$$

$$R_{Th} = \frac{1}{I_{ab}} = \frac{17 \times 6}{8} = 12.75\Omega.$$

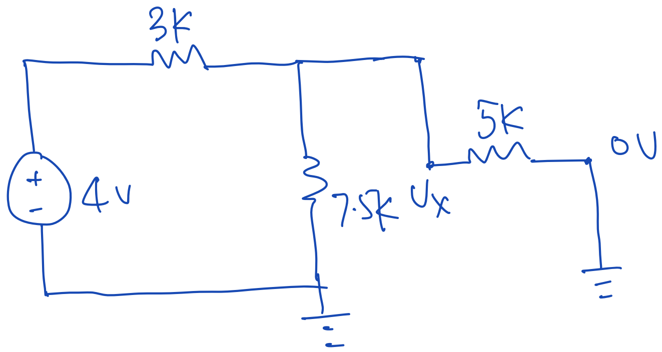
Questions 6 to 7 [4]

Consider the following op-amp circuit.



06	Find V_x [2]
07	Find i_o [2]

06



Vx can be obtained by voltage division:

$$5k \| 7.5k = \frac{5 \times 7.5}{5 + 7.5}$$

$$= \frac{5 \times 7.5}{5 \times 2.5} = 3k\Omega.$$

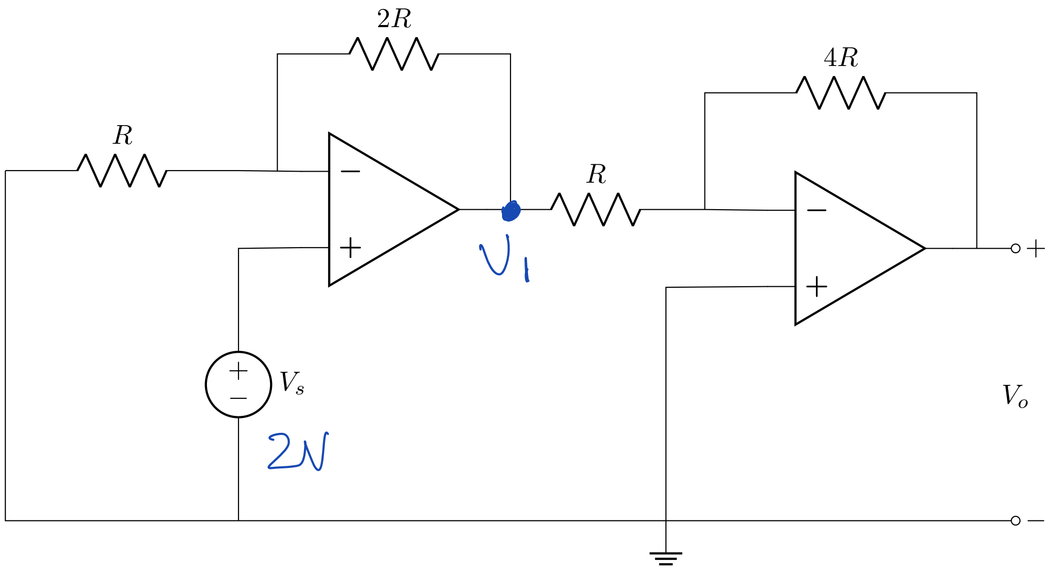
$$V_x = 2V.$$

07 summing Op Amp is observed on the right hand side.

$$V_o = -\left(\frac{10k}{4k} \cdot 2V + \frac{10k}{5k} \cdot 2V\right) = -9V$$

$$i_o = \frac{0 - (-9)}{4k\Omega} = \underline{2.25\,mA}$$

Question 8 [2]	
Consider the following op-amp circuit.	



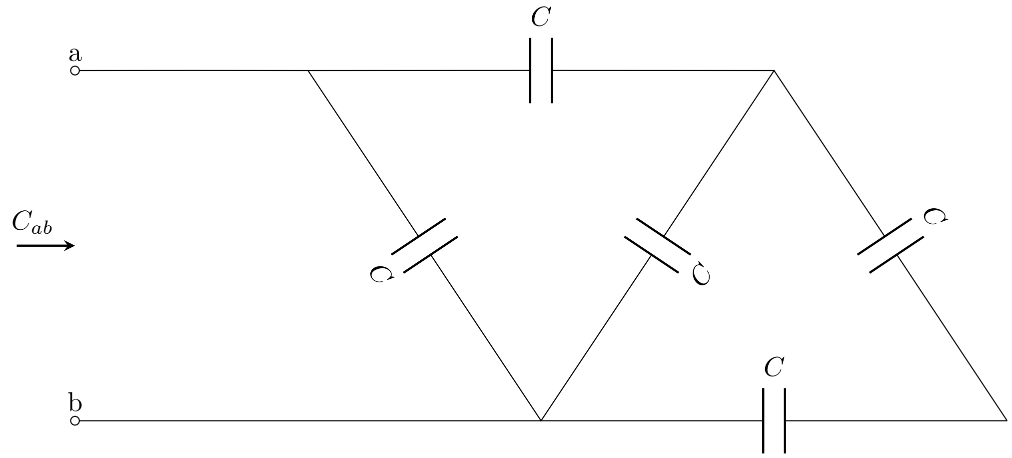
08	If $R = 10$ kΩ, $V_s = 2$ V, find V_o. [2]

$$V_1 = V_s \cdot \left(1 + \frac{2R}{R}\right) = 6V.$$

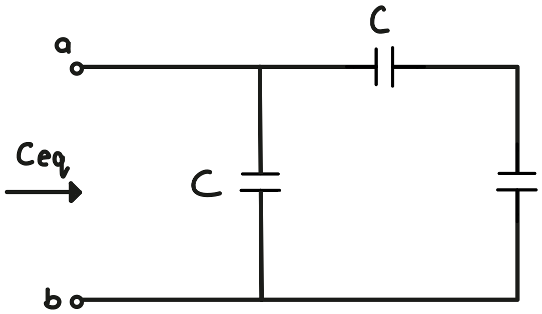
$$V_o = -V_1 \cdot \frac{4R}{R} = -24V.$$

Question 9 [2]

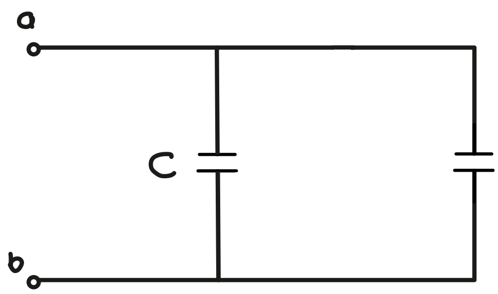
Determine the equivalent capacitance C_{ab} with respect to terminals $a - b$ if $C = 2$ F.



09	C_{ab} [2] 3.2F



$$\frac{C \times C}{C + C} + C = 1.5C$$



$$\frac{(1.5c)(c)}{1.5c + C} = 0.6C$$

$\therefore C_{ab} = 1c + 0.6c$

$C_{ab} = 1.6c$

$C_{ab} = 1.6(2)$

$C_{ab} = 3.2F$

Total Section 1: [16]

Grand Total : [16]

PRACTICE ANSWER SHEET

Assessment ID	2	0	2	4	E	B	N	1	1	1	C	0	2

	Answer	Unit
01	$R = 20$	Ω
02	$k_1 = 50$	No unit required
03	$k_2 = -20$	No unit required
04	$V_{Th} = 3$	V
05	$R_{Th} = 12.75$	Ω
06	$V_x = 2$	V
07	$i_o = 2.25$	mA
08	$V_o = -24$	V
09	$C_{ab} = 3.2$	F